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Article

Molecular Dynamics Simulation Study of the Protonation State Dependence of Glutamic Acid Transport through a Cyclic Peptide Nanotube

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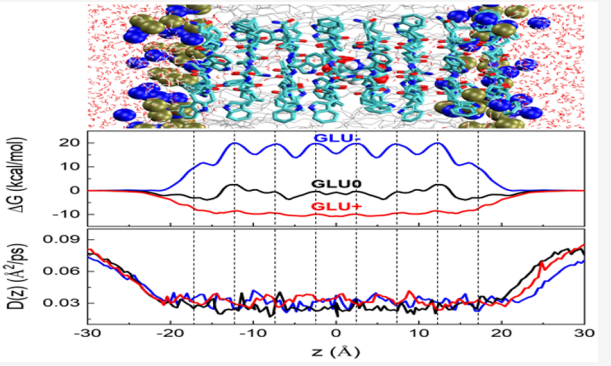
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ABSTRACT: The effect of the protonation state of glutamic acid on its translocation through cyclic peptide nanotubes (CPNs) was assessed by using molecular dynamics (MD) simulations. Anionic (GLU−), neutral zwitterionic (GLU0), and cationic (GLU+) forms of glutamic acid were selected as three different protonation states for an analysis of energetics and diffusivity for acid transport across a cyclic decapeptide nanotube. Based on the solubility-diffusion model, permeability coefficients for the three protonation states of the acid were calculated and compared with experimental results for CPN-mediated glutamate transport through CPNs. Potential of mean force (PMF) calculations reveal that, due to the cation-selective nature of the lumen of CPNs, GLU−, so-called glutamate, shows significantly high free energy barriers, while GLU+ displays deep energy wells and GLU0 has mild free energy barriers and wells inside the CPN. The considerable energy barriers for GLU− inside CPNs are mainly attributed to unfavorable interactions with DMPC bilayers and CPNs and are reduced by favorable interactions with channel water molecules through attractive electrostatic interactions and hydrogen bonding. Unlike the distinct PMF curves, position-dependent diffusion coefficient profiles exhibit comparable frictional behaviors regardless of the charge status of three protonation states due to similar confined environments imposed by the lumen of the CPN. The calculated permeability coefficients for the three protonation states clearly demonstrate that glutamic acid has a strong protonation state dependence for its transport through CPNs, as determined by the energetics rather than the diffusivity of the protonation state. In addition, the permeability coefficients also imply that GLU− is unlikely to pass through a CPN due to the high energy barriers inside the CPN, which is in disagreement with experimental measurements, where a considerable amount of glutamate permeating through the CPN was detected. To resolve the discrepancy between this work and the experimental observations, several possibilities are proposed, including a large concentration gradient of glutamate between the inside and outside of lipid vesicles and bilayers in the experiments, the glutamate activity difference between our MD simulations and experiments, an overestimation of energy barriers due to the artifacts imposed in MD simulations, and/or finally a transformation of the protonation state from GLU− to GLU0 to reduce the energy barriers. Overall, our study demonstrates that the protonation state of glutamic acid has a strong effect on the transport of the acid and suggests a possible protonation state change for glutamate permeating through CPNs.



INTRODUCTION

The cell membrane is a highly complex structure composed of various kinds of lipid molecules and membrane proteins.[1,2] The membrane separates and protects the internal compartment of a cell from the external environment by preventing hazardous foreign substances from permeating across the membrane. Due to the amphipathic nature of lipid molecules and their arrangement, the translocation of ions and hydrophilic molecules across the membrane via simple diffusion is severely disturbed.[3,4] Therefore, the flow of ions or hydrophilic molecules through membranes is assisted by specific proteins called channel proteins.[5]

Cyclic peptide nanotubes (CPNs) are a family of synthetic channel assemblies that are formed through intermolecular hydrogen bonding between cyclic peptide rings.[6−9] The cyclic peptide ring in CPNs consists of alternating D- and L-type amino acid residues, and the side chains and the number of amino groups in a ring determine the hydrophilic or hydrophobic

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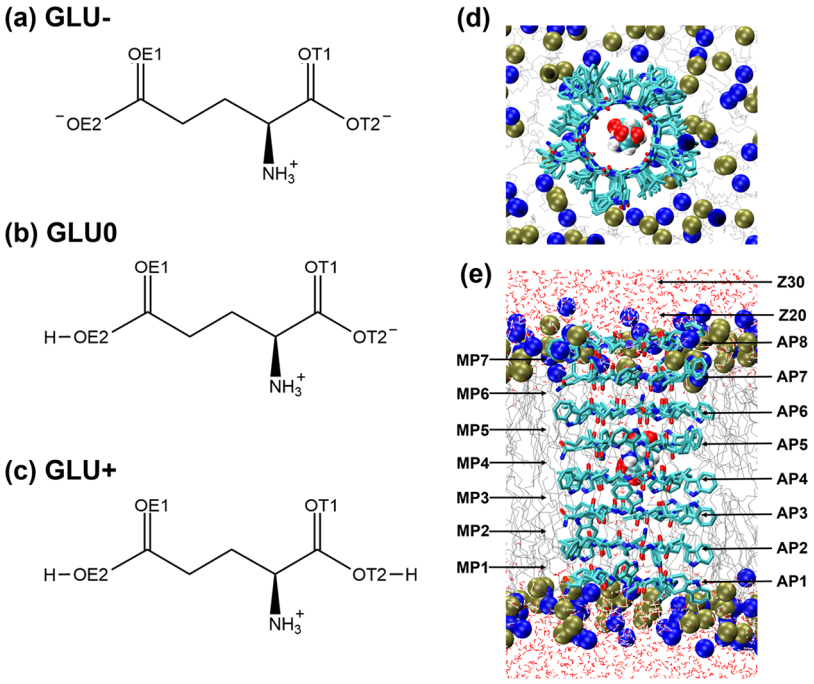


Figure 1. Structures of (a) anionic (GLU−), (b) zwitterionic (GLU0), and (c) cationic (GLU+) forms of L-glutamic acid and (d) top and (e) side views of a cationic form of glutamic acid permeating through a cyclic peptide nanotube (CPN), 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] embedded in a DMPC lipid bilayer. In (d) and (e), hydrogen atoms of the CPN and DMPC are eliminated for clarity, lipid tails are represented by gray tubes, and P and N atoms in the lipid head are indicated as tan and blue spheres, respectively. In (e), AP means the α-plane regions which are located at ±2.45, ±7.35, ±12.25, and ±17.15 Å in the z direction and MP means the midplane regions positioned between the two neighboring cyclic peptide rings. The Z20 and Z30 regions are defined as the position of glutamic acid at z = 20 and z = 30 Å, respectively. The Z20 region corresponds to the position at the channel entrance of the CPN.

nature and the diameter of the synthetic channel, respectively.[10−12] Compared with other types of nanotubes such as carbon nanotubes, CPNs have high biocompatibility because they are made up of only amino acids which are generally used as a vital element in the human body, and as a result, many applications have been proposed including as an anticancer drug delivery system and as an antibacterial agent.[13,14] Like pore-forming natural channel proteins, CPNs run through the cell membrane and the lumen of the CPNs acts as a passage for ions or hydrophilic small molecules.[7,15] Specifically, in the case of cyclic decapeptide nanotubes where the cyclic peptide ring is composed of ten alternating D- and L-amino acid moieties, they can mediate the transport of small drug agents and biomolecules such as 5-fluorouracil, glucoses, and glutamic acids across membranes.[14−17] For example, combining experimental measurements and computational approaches, Liu et al. demonstrated that cyclic decapeptide nanotubes can serve as a drug delivery system for 5-fluorouracil compound, an anticancer drug.[14,18,19] D-Glucose permeation across membranes facilitated by cyclic decapeptide nanotubes was also studied experimentally and computationally.[15,20] L-Glutamic acid is one of the most common amino acids that plays an important role in protein synthesis and cellular metabolism.[21] The acid can take one of three protonation states, including anionic (GLU−), zwitterionic (GLU0), and cationic (GLU+) forms, depending on the environmental pH. L-Glutamate, the anionic form of L-glutamic acid, plays a key role as an excitatory neurotransmitter in the nervous system, and malfunctioning of glutamate transporters causes neurodegenerative disorders including Alzheimer's disease.[22−26] Above pH 4.1, glutamic acid has both carboxylic groups deprotonated and only the amine moiety protonated, becoming a negatively charged glutamate (Figure 1a).[27] At pH between 2.1 and 4.1, the carboxyl group in the side chain gains a proton, and the molecule takes a neutral zwitterionic form (Figure 1b). Under sufficiently acidic conditions, the other carboxylate that is close to the amine group is also protonated, and the acid carries a net positive charge (Figure 1c). Sánchez-Quesada and co-workers experimentally demonstrated that glutamic acid could be translocated from the inside to the outside of large unilamellar vesicles mediated by cyclic decapeptide nanotubes. They measured the biionic and reverse potentials for NaCl and NaGlu currents through gramicidin and cyclic octa- and decapeptide scaffolds embedded in planar lipid bilayers and obtained permeability ratios of ca. 6.4 for sodium and glutamate ions using the Goldman−Hodgkin−Katz (GHK) equation.[5,28]

The MD simulation method is a powerful computational tool that can be used to explore complex biological systems at the atomistic level.[29,30] Based on the principles of statistical mechanics, the method enables us to elucidate the underlying mechanisms of various biophysical processes. In the case of CPNs, MD simulations have been extensively utilized to examine the transport behavior of small biomolecules or organic molecules through the synthetic channel protein spanning across the lipid bilayers. For example, the permeation of biomolecules such as lactic acid and glucose through cyclic decapeptide nanotubes was scrutinized using MD simulations, which revealed that the energetic and dynamic properties of the molecules inside the CPNs depend on their isomeric composition.[20,31]

In this paper, we used all-atom MD (AAMD) simulations to examine the influence of the protonation state of glutamic acid on its transport through a cyclic decapeptide nanotube. In order to obtain the energetics and dynamic properties of the different

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charge states of the acid inside the CPN, the potential of mean force (PMF) and position-dependent diffusion coefficient profiles were calculated for the anionic, neutral zwitterionic, and cationic forms of glutamic acid. Hydrogen bond formation and radial distribution functions were also determined for the three charge states of the acid inside the CPN. Based on the solubility-diffusion model,[32−35] we evaluated permeability coefficients for the three states using the calculated PMF and diffusivity profiles and compared the results with those obtained from the experiments conducted by Sánchez-Quesada and co-workers. From the comparison, we proposed several possibilities in the glutamate permeation mediated by CPNs.

The paper is organized as follows. The simulation system and computational details are introduced in the next section. The calculated results are given in the Results and Discussion, where comparisons with experimental measurements are also considered. In the final section, conclusions and further work related to the current study are addressed.

COMPUTATIONAL METHODS

Description of Simulations for Transport of Three Protonation States of Glutamic Acid through a 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] Nanotube. The model systems are composed of a single decapeptide CPN, 98 DMPC (1,2-dimyristoyl-*sn*-glycero-3-phosphocholine) lipid molecules, 4717 water molecules, and a single anionic (GLU−), zwitterionic (GLU0), or cationic (GLU+) form of L-glutamic acid in a simulation box whose dimension is about 50 Å × 55 Å × 91 Å (Figure 1d and e). The decapeptide CPN used in this study is made of 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] peptide rings. The channel axis of the CPN that is normal to the DMPC bilayer surface was set up to be aligned parallel to the z axis of the system. Simulations were conducted by using the NpT ensemble with periodic boundary conditions applied in all directions. Simulation systems were fully equilibrated at 303 K and 1 bar using the Langevin thermostat and the Nosé−Hoover Langevin piston method.[36] The damping coefficient, γ_L in the Langevin thermostat, which substantially affects the diffusion coefficient of glutamic acid in a simulation, was selected to be 3.0 ps^{-1} because a diffusion coefficient of 0.081 Å^2/ps was obtained with that value of γ_L from the mean-square displacement (MSD) analysis combined with an MD simulation and is close to an experimental value of 0.073 Å^2/ps.[37] Electrostatic interactions were evaluated using the particle mesh Ewald (PME) method, while van der Waals interactions were truncated at 12 Å and were gradually switched to 0 from 10 to 12 Å. The CHARMM22 force field set was applied for the CPN and the three forms of glutamic acid.[38] The CHARMM36 force field was used for DMPC lipid molecules, and the TIP3P water model was employed for water.[39,40] All the MD simulations were performed using NAMD 2.13, and visualizations were realized by VMD 1.9.0.[41,42]

Construction of PMF and Position-Dependent Diffusion Coefficient Profiles from a Series of Simulation Trajectories. Symmetrized PMF profiles for the three protonation states of glutamic acid were calculated as a function of the z coordinate of the center of mass (CoM) of each state of the acid based on the adaptive biasing force (ABF) and thermodynamic integration (TI) methods.[43] For the PMF calculations, the channel axis of the CPN was maintained parallel to the z direction using the colvar method. A spring constant of 25 kcal/mol/Å^2 was used to restrain the CoM of C_α atoms of a peptide ring in the corresponding α-plane region and to keep the distance between the two neighboring peptide rings at 4.9 Å in the z direction. The divergence issue due to the integration over all xy areas outside the CPN was avoided by applying a weak harmonic constraint with a spring constant of 0.16 kcal/mol/Å^2 to each protonation form of glutamic acid that keeps the CoM of each form within 1.5 Å of the channel axis in the xy plane. The reaction coordinates of each protonation form of glutamic acid range from −30.0 to 30.0 Å in the z direction and were divided into 12 sections. In each section two independent 200 ns MD simulations were carried out and the sectional PMF was evaluated using the ABF method with a bin width of 0.4 Å. The acquired sectional PMF profiles were combined and symmetrized to construct the complete PMF profiles for the three protonation states of glutamic acid. The contributions of system components (CPN, water, and DMPC bilayer) to the PMF profiles of the protonation states were analyzed by decomposing the free energy using the TI method. The mean forces acting on different protonation forms of glutamic acid from each component were calculated with trajectories of ABF simulations and integrated to construct separated energy profiles. The contributions of system components such as CPN, water, and DMPC bilayer to the PMF profiles of the protonation states were analyzed by decomposing the free energy using the TI method. The mean forces acting on the different protonation forms of glutamic acid from each component were calculated using the trajectories of ABF simulations and integrated to construct decomposed free energy profiles.[20] Based on the Bayesian inference (BI) method combined with a Monte Carlo (MC) technique, position-dependent diffusion coefficients of the three protonation states of glutamic acid, $D(z)$, were calculated as a function of the z coordinate of the CoM of each form to examine the protonation state dependence of the dynamic behavior of glutamic acid inside the CPN.[44,45] A simple diffusion model fails to describe the diffusive behavior of a permeant inside a CPN due to complicated interactions of the permeant with system components such as channel waters. Instead, the diffusion process is well accounted for by a subdiffusion model and the calculation of position-dependent diffusion coefficients of glutamic acid is affected by the observation time or lag time, Δt, used in the BI/MC method.[46−48] In our study, we set the value of Δt to be 2 ps for calculations of the position-dependent diffusion coefficient according to a previous study of glucose transport through a CPN.[20] The ABF trajectories for the PMF calculations were employed as input coordinates for the BI/MC method. The position-dependent diffusion coefficient profiles for the three protonation states were reconstructed using a fitting function given as[49]

$$D^{\mathrm{fit}}(z) = D_0\left(a_1 + \frac{1.0 - a_1}{1.0 + \mathrm{e}^{-(|z|-z_0)/\Delta z}}\right) \quad (1)$$

where D_0 is the bulk diffusion coefficient of a charge state of glutamic acid and a_1, z_0, and Δz are fitting parameters whose values are provided in Table S1. The fitted position-dependent diffusion coefficient profiles for the three charged states of glutamic acid are represented in Figure S13.

RESULTS

Figure 2 represents the symmetrized PMF profiles constructed using the ABF and TI methods for the three charge states of glutamic acid permeating through the CPN with DMPC lipid bilayers in the z direction. The PMF profiles obtained from the

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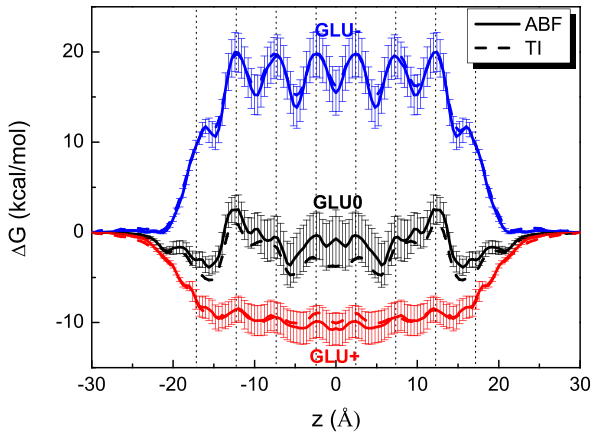


Figure 2. Symmetrized PMF profiles with error bars for three charge forms of glutamic acid across the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers calculated via the ABF and TI methods. The vertical broken lines indicate the position of each cyclic peptide ring in the z direction.

ABF method are in good agreement with those from the TI method except for a slight difference in the case of the zwitterionic form (GLU0). The dependence of free energy variation on the protonation state of glutamic acid is clearly shown in Figure 2. As glutamic acid becomes deprotonated or negatively charged, energy barriers are developed inside the CPN due to the cation-selective nature of the lumen of the CPNs.[50] Several local energy barriers and wells observed in the PMF curves suggest a hopping mechanism for glutamic acid diffusion inside the CPN regardless of the protonation state. The PMF comparisons between the three protonation states of glutamic acid and with Na^+ and Cl^- in the same CPN illustrate that GLU+ and GLU− have slightly lower energy minima and higher energy maxima than Na^+ and Cl^- do, respectively (Figure S1). This can be attributed to the larger size of glutamic acid than the ions, leading to greater interactions of glutamic acid with the lumen of CPNs. As for GLU− and GLU+, energy wells and barriers are placed at the midplane and α-plane regions inside the channel, respectively, while those for Na^+ and Cl^- are located oppositely, which can also be attributed to the size difference between glutamic acid and the ions.

To analyze the origin of these distinct features in the PMF profiles depending on the protonation state of glutamic acid, the PMF profiles were decomposed into contributions from system components, such as CPN, membrane, and water, and the characteristic decomposition profiles are illustrated in Figure 3. As the anionic form, GLU−, enters into the lumen of the CPN, it is stabilized by water but is seriously destabilized by the DMPC bilayer, probably due to electrostatic repulsions with the negative charge bearing carbonyl oxygens of the acyl groups in the tail regions of DMPC bilayers (Figures 3a, S2a, and S3a). Although the CPN slightly contributes to the free energy barriers through the channel for GLU−, favorable interactions between GLU− and the CPN are also observed just outside the channel entrances. These favorable interactions of GLU− with the CPN can be ascribed to electrostatic attractions between the negatively charged carboxyl groups of GLU− and positively charged free amide hydrogens in the outermost cyclic peptide rings of the CPN (Figure S3a). Destabilization by the bilayer and CPN is more dominant inside the CPN, leading to the overall energy barriers in the total PMF curve for GLU−. The PMF decomposition profiles for GLU− are similar to those for Cl^-

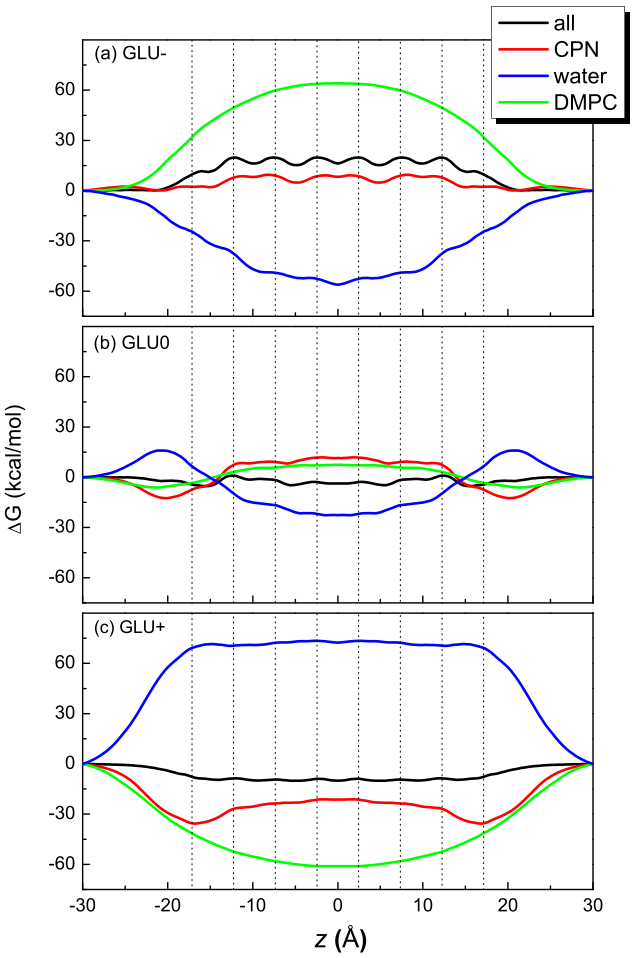


Figure 3. Symmetrized PMF decomposition profiles for (a) GLU−, (b) GLU0, and (c) GLU+ along the z axis. The profiles were evaluated from the ABF simulation trajectories using the TI method. The vertical broken lines are the same as those in Figure 2.

except that Cl^- has stronger interactions with the system components than GLU− due to the smaller size of Cl^- (Figure S5a). In contrast, as GLU+ comes through the lumen of CPN, the cationic form is stabilized by the DMPC bilayer and CPN, but it is destabilized by water (Figure 3c). Although the destabilization caused by water is huge, the total free energy becomes negative inside the CPN due to strong electrostatic interactions of the cationic form with the negatively charged carbonyl groups of CPNs and the acyl groups of the DMPC bilayers (Figure 3c). Similar energetic behavior for permeation through the CPN is also observed for Na^+ (Figure S5b). Interestingly, in the case of zwitterionic glutamic acid, it exhibits mixed patterns with the other forms in Figure 3b. Like the cationic form, GLU0 is stabilized by the lipid and CPN and destabilized by water outside the CPN, but these features are reversed as it enters into the channel. These conflicting behaviors of the zwitterionic glutamic acid inside and outside the CPN can be attributed to the fact that GLU0 has both a protonated and deprotonated carboxyl group while the two carboxyl groups are totally deprotonated and protonated in GLU− and GLU+, respectively.

Because one of the key factors affecting the energetics of glutamic acid translocation is hydrogen bonding between glutamic acid and the system components, we calculated the number of hydrogen bonds that each type of glutamic acid forms

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with water molecules, the CPN backbone, and the glutamic acid itself. Glutamic acid has four hydrogen bond acceptor sites (OE1, OE2, OT1, OT2) regardless of protonation state, but the number of hydrogen bond donor sites depends upon the protonation state as listed in Table 1. In addition to water molecules, the three protonation states of glutamic acid can form hydrogen bonds with the carbonyl oxygens and amide hydrogens in the backbone of the CPN as the glutamic acid passes through the CPN. The formation of hydrogen bonding was determined according to the definition by Luzar and co-workers where a hydrogen bond is considered to be formed when the interoxygen distance, R_{OO} < 3.5 Å, and, at the same time, the angle between the O−O axis and the O−H bond has θ_{HOO} < 30° between two molecules or inside a molecule.[51] As shown in Figure 4a, the total number of intermolecular and intramolecular hydrogen bonds of a glutamic acid decreases inside the CPN due to the confinement imposed by the lumen of the CPN, although new hydrogen bonds with the backbone of the CPN are created. It is also observed in Figure 4a that GLU−, which carries one positively charged amino group and two negatively charged carboxyl groups, builds the largest number of hydrogen bonds both inside and outside the CPN while GLU+ forms the smallest number of hydrogen bonds, owing to only one positively charged amino group. Figures 4b and S7a show that inside the CPN GLU− forms hydrogen bonds with water molecules better in the α-plane regions (AP regions) than in the midplane regions (MP regions) because GLU− in the AP regions interacts with water molecules in the MP regions where more water molecules are found than in the AP regions.[52] Although GLU− plays a minor role as a hydrogen bond donor, it serves as a strong hydrogen bond acceptor and forms the largest number of hydrogen bonds with water due to the four negatively charged deprotonated carboxyl oxygens (Figure S7b). It also explains why the anionic form has favorable interactions with water molecules inside the CPN in Figure 3a. Compared with the neutral or anionic form of glutamic acid, GLU+ develops the largest number of hydrogen bonds with water as a donor inside and outside the CPN because the cationic form has the largest number of donor sites, as indicated in Table 1 (Figure S9a). On the other hand, GLU+ as a hydrogen bond acceptor has the smallest number of hydrogen bonds with water because the acceptor sites in the cationic state are completely protonated and the attached hydrogens will interfere with hydrogen bonding between the carboxyl oxygens of GLU+ and water hydrogens. Figure 4c illustrates that the hydrogen bonds between GLU− and CPN are more prominent in the MP regions rather than in the AP regions because the hydrogens of the protonated amide group NH_3^+ in GLU− can interact and form hydrogen bonds with the carbonyl oxygens of the CPN backbone in the MP regions. Because the carbonyl oxygens in the backbone of the CPN carry negative charges, the NH_3^+ groups of GLU− are expected to form hydrogen bonds with those carbonyl oxygens in the MP regions (Figure S10a). Compared with GLU−, GLU0 and GLU+ as a donor create more hydrogen bonds with the CPN at the AP regions near the channel entrances due to the greater number of hydrogen donor sites (Table 1 and Figure S10a). Because most amide hydrogens in the backbone of the CPN are involved in the hydrogen bonds with the carbonyl oxygens in the adjacent cyclic peptide rings except for the free amide hydrogens at the outermost rings, glutamic acid as an acceptor can form hydrogen bonds only with the dangling amide hydrogens at the outermost rings of the CPN regardless of its protonation state, as shown in Figure S10. In the case of GLU−, the formation of intramolecular hydrogen bonds is observed

Table 1. List of Hydrogen Bond Donor and Acceptor Sites in the Three Charge States of Glutamic Acid

Charge state	GLU−	GLU0	GLU+
Donors	3×HN	3×HN, HOE2	3×HN, HOE2, HOT2
Acceptors	OE1, OE2, OT1, OT2	OE1, OE2, OT1, OT2	OE1, OE2, OT1, OT2

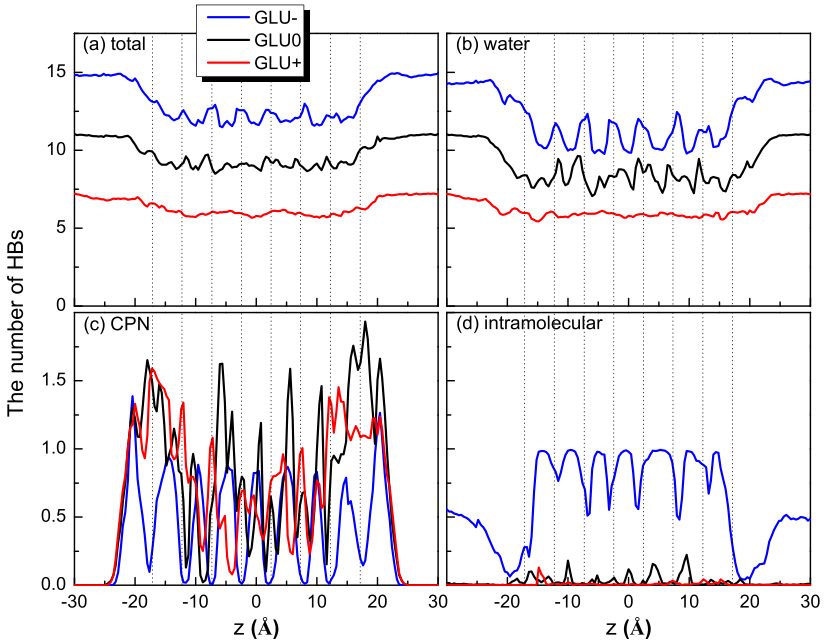


Figure 4. Number of hydrogen bonds formed by each protonated state of glutamic acid with (a) all the components, (b) water, (c) CPN, and (d) itself along the z axis. The vertical broken lines are the same as those in Figure 2.

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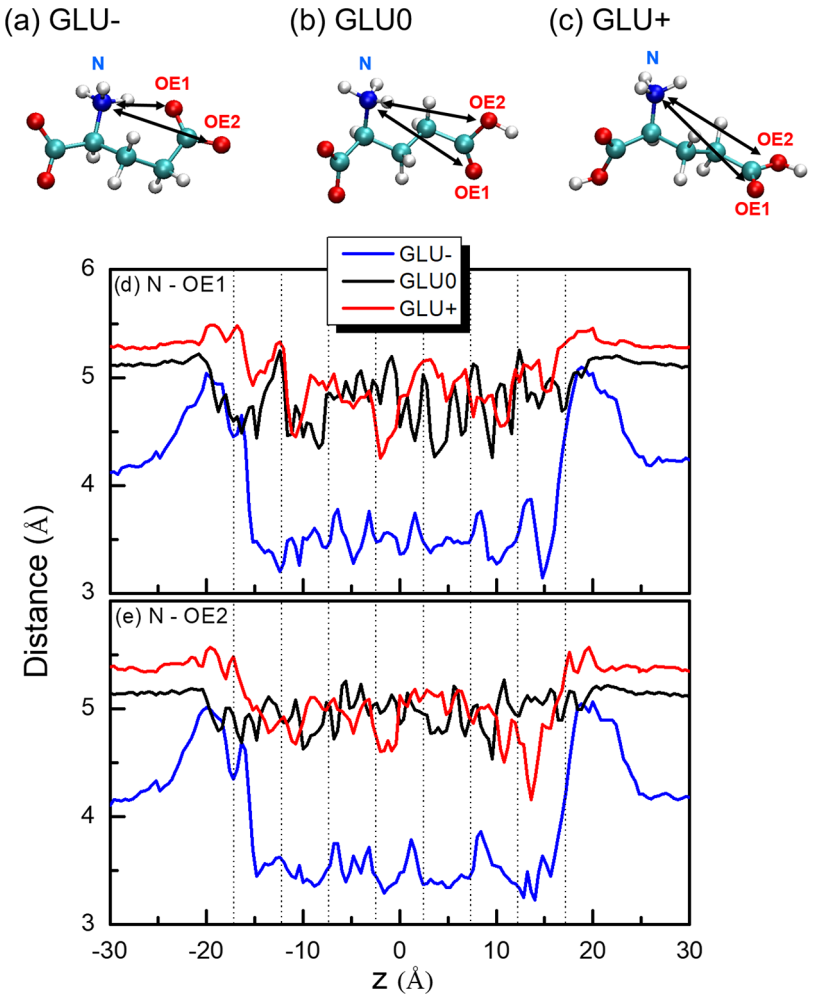


Figure 5. Typical structures and definitions of the distance between the nitrogen atom (N) in the protonated amide group and the oxygen atoms (OE1 and OE2) in the carboxyl or carboxylate group of the side chain in (a) GLU−, (b) GLU0, and (c) GLU+ and distance profiles (d) between N and OE1 and (e) between N and OE2 for the three protonation states of glutamic acid.

inside and outside the CPN in Figure 4d. In order to examine the possibility of intramolecular hydrogen bonding of the three protonation states of glutamic acid, we calculated the distance between the nitrogen atom (N) of the NH_3^+ group and the two oxygen atoms (OE1 and OE2) of the side chain COO^- (or COOH) group as a function of z for the CoM of each protonation state (Figure 5a−c). Parts d and e of Figure 5 clearly demonstrate that the distances between N and OE1 and OE2 for GLU− are much smaller than those for GLU0 and GLU+ inside as well as outside the CPN, indicating that the anionic form prefers developing intramolecular hydrogen bonds inside the CPN due to avoiding unfavorable interactions with the lumen of the CPNs as shown in Figure S3a. The rapid increase of the distance at a position of $z = \pm 20.0$ Å for GLU− suggests that the formation of intramolecular hydrogen bonds is interfered by the dangling carbonyl oxygens and amide hydrogens in the outermost cyclic peptide rings of the CPN. Indeed a cyclical hydrogen-bonded structure of GLU− was proposed in an activity measurement experiment for sodium glutamate in bulk water.[53]

To explore hydrogen bonds and interactions between glutamic acid and water molecules, radial distribution functions (RDFs) of water hydrogen atoms around the three protonation forms of glutamic acid were calculated and are shown in Figure 6. Inside the CPN, the three amide hydrogens (three HNs) of GLU− have a greater chance to find and form hydrogen bonds with water molecules in the AP regions than in the MP regions (Figure 6a and b), and the peak heights of the RDFs in the Z20 (z = 20 Å) region are very similar to those in the AP6 regions (Figure 6c). In addition, Figures 6a−c and S7a−c show that GLU− serves as a hydrogen bond acceptor better than a donor. The RDF profiles for GLU− in Figure 6a−c show good agreement with the hydrogen bond number calculations in Figure 5 where more hydrogen bonds are formed between GLU− and water molecules in the AP region than in the MP regions. In the case of GLU0, the three HNs and the carboxyl hydrogen atom of the side chain (HOE2) have a similar number of water molecules around them in the MP4 and Z20 regions, but in the AP6 region, more water molecules are found around the three HNs than in the HOE2 (Figure 6d−f). The RDF profiles for GLU0 imply that in the MP and Z20 regions the three HNs and HOE2 of GLU0 are involved in the formation of hydrogen bonds with water, but in the AP regions, the carboxyl hydrogen of the side chain mainly participates in the hydrogen bonding with the carbonyl oxygens of the CPN, leading to reduced interactions with water. Parts g and h of Figure 6 illustrate that the three HNs and HOE2 of GLU+ are strongly interacting and forming hydrogen bonds with water molecules in the MP4 region, but the other carboxyl hydrogen, HOT2, weakly joins hydrogen bonding with water resulting from strong

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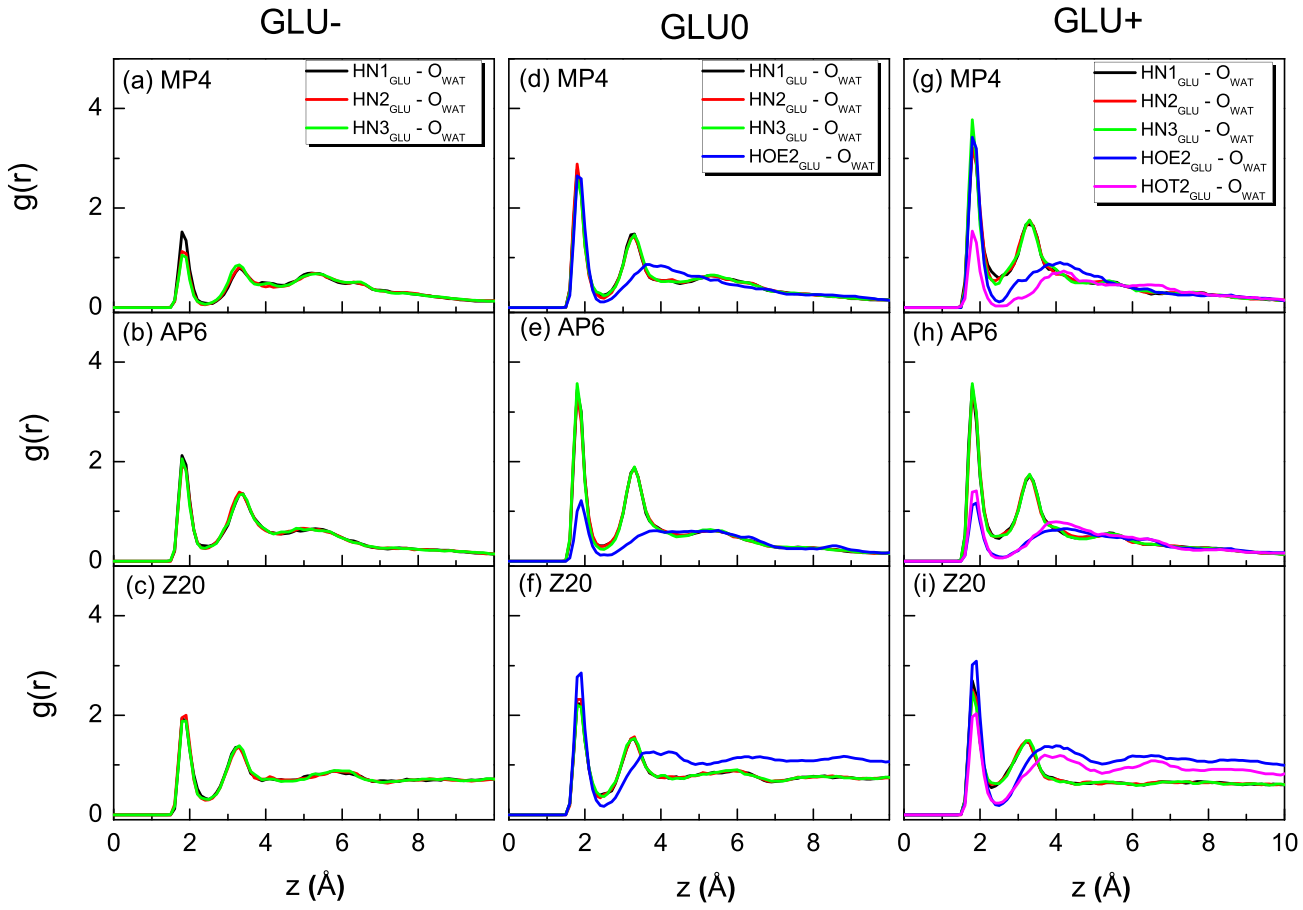


Figure 6. RDF profiles for water oxygen atoms around the hydrogen atoms of (a–c) GLU−, (d–f) GLU0, and (g–i) GLU+ in the MP4, AP6, and Z20 regions. The MP4, AP6, and Z20 regions are defined in Figure 1e.

interactions and hydrogen bonds with the carbonyl oxygens of the CPN backbone. In the AP6 region, only the three HNs of GLU+ are mainly involved in interactions with water molecules. The RDF profiles for GLU+ near the channel entrances (Z20 region) appear similar to those in the MP4 region, but the peak heights are slightly smaller probably due to the competition between water oxygens and free carbonyl oxygens of the outermost cyclic peptide rings. When the carboxyl groups of GLU0 and GLU+ are protonated, their role as a hydrogen bond acceptor is drastically reduced as shown in Figure S7d–i.

In Figure 7a–c, RDF profiles for the carbonyl oxygens in the CPN backbone around the amide hydrogens of GLU− indicate that GLU− as a hydrogen bond donor can build hydrogen bonds with the carbonyl oxygens of the CPN in the MP regions and the channel entrances but fails to form hydrogen bonds with the carbonyl oxygens in the AP regions. These RDF results are consistent with the hydrogen bond number calculations between GLU− and the CPN in Figure 4c where the number of hydrogen bonds is close to one in the MP regions but the number becomes negligible in the AP regions. The three HNs and HOE2 of GLU0 weakly interact and form hydrogen bonds with the carbonyl oxygens of the CPN in the MP regions, while the carboxyl oxygen of the side chain of GLU0 has a strong interaction and forms hydrogen bonds with the carbonyl oxygens of the CPN backbone in the AP regions (Figure 7d and e). As shown in Figure 7f, at the channel entrances, interactions of GLU0 with the carbonyl oxygens of the CPN appear analogous to those at MP4. The low peak heights with the CPN for GLU0 in the MP4 region and channel entrances in Figure 7d and f indicate that strong interactions with water in those regions interrupt interactions between the hydrogens of GLU0 and the carbonyl oxygens of the CPN. For GLU+ in the MP4 region, only the carbonyl oxygen of the α carbon, HOT2, interacts and generates hydrogen bonds with the carbonyl oxygens of the CPN, but in the AP6 region, both the carboxyl oxygens of the α carbon and side chain are strongly involved in the interactions and hydrogen bond formations with the carbonyl oxygens in the CPN backbone (Figure 7g and h). As shown in Figure 7i, the three amide hydrogens and the α carbon hydrogen of GLU+ participate in the interactions with the carbonyl oxygens of the CPN at the channel entrances. Although the hydrogens in the amide and carboxyl groups of glutamic acid actively participate in interactions and form hydrogen bonds with the carbonyl oxygens of the CPN as well as water oxygens, the amide hydrogens in the CPN backbone mainly form hydrogen bonds with the carboxyl oxygens of the CPN in the MP or AP regions, except for the channel entrances where they develop hydrogen bonds with the three forms of glutamic acid to some extent (Figure 7a–i).

Figure 8 illustrates representative configurations of the three protonation states of glutamic acid taking part in hydrogen bonds with the CPN backbone and water molecules. In Figure 8a and b, cyclical structures for GLU− through an intramolecular hydrogen bond are clearly seen in the AP and MP regions, and indeed, this kind of cyclical structure was proposed in an experiment with sodium glutamate.[53] In the AP6 regions,

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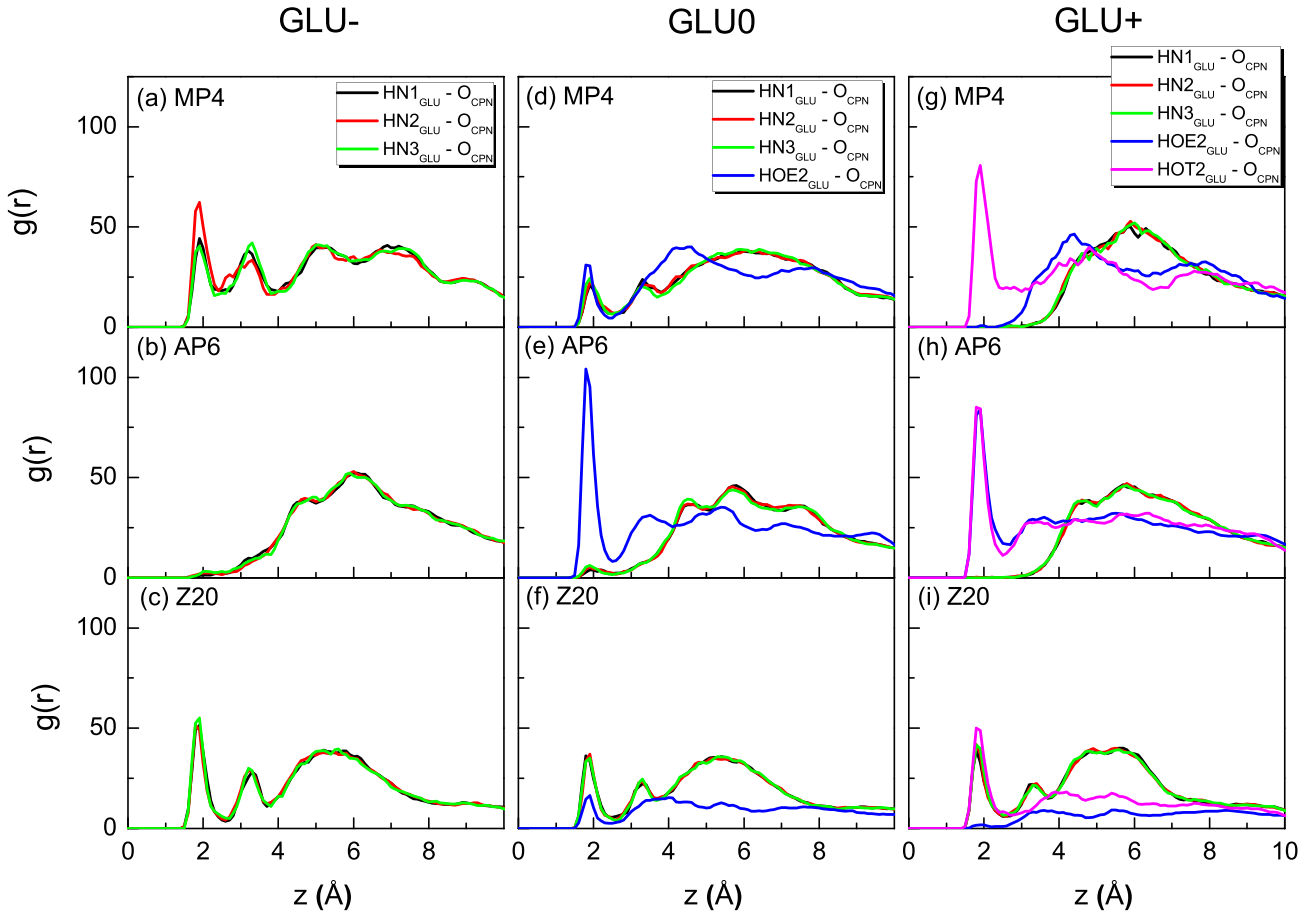


Figure 7. RDF profiles for the carbonyl oxygen atoms of the CPN backbone around the hydrogen atoms of (a–c) GLU–, (d–f) GLU0, and (g–i) GLU+ in the MP4, AP6, and Z20 regions. The definitions of the regions are the same as those in Figure 6.

the three protonation states generally have elongated structures along the channel axis and form hydrogen bonds with water molecules in the adjacent MP and AP regions (in Figure 8a, c, and e). On the other hand, the three forms of glutamic acid can generate a larger number of conformations in the MP regions and can build hydrogen bonds not only with water in the same MP region and adjacent AP and MP regions but also with the carbonyl oxygens of the CPN backbone in the same MP region (in Figure 8b, d, and f).

Position-dependent diffusion coefficient profiles for the three protonation states of glutamic acid along the CPN z axis are presented and compared with those in bulk water in Figure 9. The diffusion coefficients of all the protonation states of the acid begin to decrease more than 10 Å away from the channel entrance, indicating that changes in the interaction between glutamic acid and surrounding water molecules occur much earlier than the glutamic acid arrives at the entrance of the CPN. Similar diffusivity behaviors were also observed for glucose molecules transporting through the CPN.[20] In Figure 9 the diffusion coefficients for the three protonation forms in bulk water decrease from 0.093 through 0.081 to 0.077 Å^2/ps as the number of protonation sites in glutamic acid increases from GLU+ to GLU–. This can be attributed to the fact that, as the number of charged sites in glutamic acid increases, the local interactions and hydrogen bonds of glutamic acid with water molecules become stronger, leading to an increase in the hydrodynamic radius and the reduction of the diffusion coefficient of the acid. However, the three forms of glutamic acid have comparable diffusion coefficients inside the CPN regardless of the protonation state. The fluctuating part of the force is governed by the local interaction between a protonation state of the acid and channel water molecules, and due to confinement imposed by the lumen of CPNs, all the three protonation states of glutamic acid have similar local interactions with the confined channel water. As a result, the frictional behaviors of the three states are alike.[49,54] The position-dependent diffusion coefficient profiles for the three protonation states were reconstructed using a fitting function given, as indicated in eq 1. These fitted diffusivity functions, $D^{\mathrm{fit}}(z)$, were employed to calculate the local resistivity of the three protonation states of glutamic acid.

DISCUSSION

Based on the solubility-diffusion model,[33,34,55,56] the permeability coefficient for a molecule or ion i through a CPN, P_i^{CPN}, can be expressed as

$$P_i^{\mathrm{CPN}} = \left[\int_{z_1}^{z_2} \mathrm{d}z R_i^{\mathrm{CPN}}(z)\right]^{-1} \tag{2}$$

where $z_2 = -z_1 = L/2$ where L is the box length in the z axis and $R_i^{\mathrm{CPN}}(z)$ is the local resistivity depending on the z coordinate of i, which is given as

$$R_i^{\mathrm{CPN}}(z) = \frac{\exp[\beta \Delta G_i(z)]}{D_i^{\mathrm{fit}}(z)} \tag{3}$$

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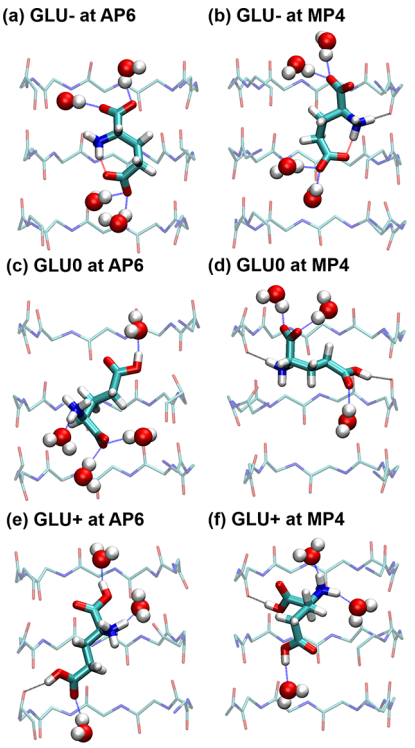


Figure 8. Representative configurations of the three protonation states of glutamic acid interacting with the CPN backbone and channel water molecules in the AP6 and MP4 regions for (a and b) GLU−, (c and d) GLU0, and (e and f) GLU+. Based on the definition of hydrogen bonding formation in the Computational Methods section, hydrogen bonds between donors and acceptors are shown with broken lines in which hydrogen bonds with the CPN and water are colored in black and blue, respectively, and intramolecular hydrogen bonds are in red. For clarity, side chains of the CPN were removed and only some of all the water molecules forming hydrogen bonds with the protonation states are displayed.

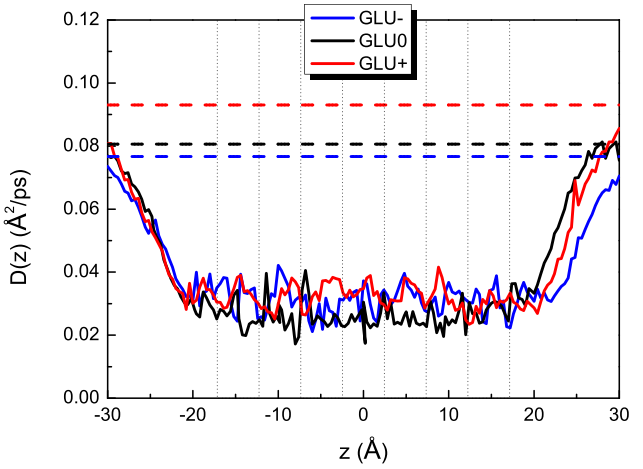


Figure 9. Position-dependent diffusion coefficient, $D(z)$, profiles for the three protonation states of glutamic acid along the z axis. The horizontal broken color lines represent the diffusion coefficient of each protonation state of glutamic acid in bulk water acquired from the MSD analysis. The vertical broken lines are the same as those in Figure 2.

Here $\beta = 1/k_BT$ where k_B is the Boltzmann constant and T is the temperature and $\Delta G_i(z)$ and $D_i^{fit}(z)$ are the PMF and fitted position-dependent diffusivity for i, respectively. Combining the energetics and dynamics of permeating particle i, $R_i^{CPN}(z)$ describes the local transport behavior of the particle through a CPN. Figure 10 illustrates the local resistivity $R_i^{CPN}(z)$ for the

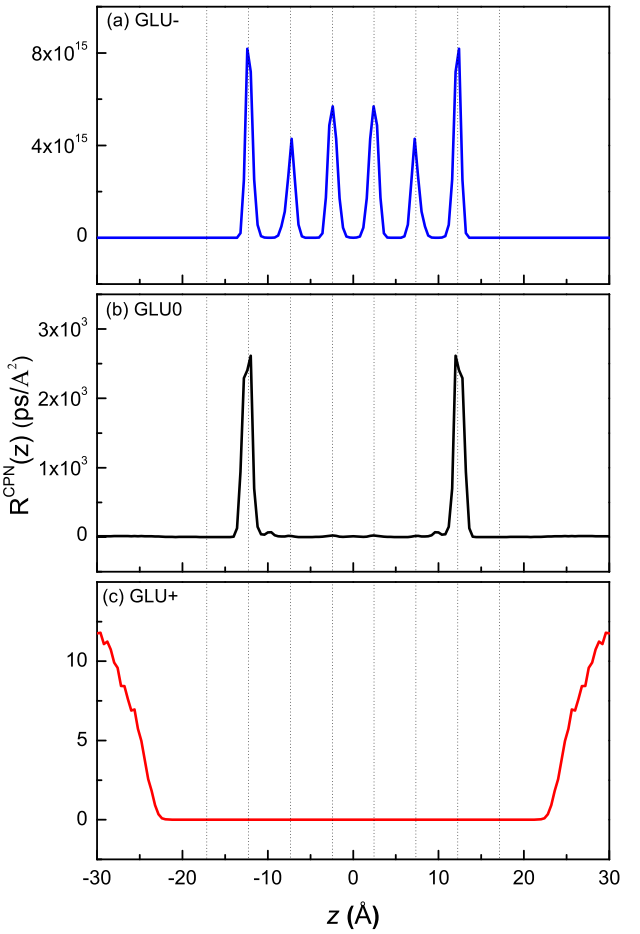


Figure 10. Local resistivity, $R_{CPN}(z)$, profiles for the three protonation states of glutamic acid along the z axis. The fitted functions for $D(z)$ were used for calculations of $R(z)$ in eq 3. The vertical broken lines are the same as those in Figure 2.

protonation states of glutamic acid. Compared with GLU+ and GLU0, GLU− yields enormous local resistivities inside the CPN as the anionic glutamic acid (or glutamate) permeates through the CPN. Calculated via eq 2, the permeability coefficients for the three protonation states are tabulated in Table 2. The permeability coefficients show the strong dependence of the coefficient on the energetics rather than the diffusivity of each species due to the substantial difference in PMF energetics and similar local diffusivity of the three states. Consequently, the extremely small permeability coefficient for GLU− inside the CPN indicates that glutamate is unlikely to translocate through the CPN. Additionally, given the numbers in Table 2, the calculated permeability ratio for Na^+ and GLU−, $\chi^{calc} = P^{calc}_{Na^+}/P^{calc}_{GLU-}$, is 4.01×10^{14}, which is by far larger than an experimentally estimated ratio, $\chi^{exp} = 6.4$, from the biionic and reverse potential measurements for NaCl and NaGlu currents through the CPN embedded in a planar lipid bilayer.[16] To elucidate the discrepancy between the simulation results and the experimental measurements, we propose several possibilities. The first possibility is that, because the flux of a molecule is given as $J = P_m\Delta C$, where J is the flux, P_m is the permeability coefficient, and ΔC is the concentration difference of the

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Table 2. Permeability Coefficients, P_{CPN}, of the Three Charge States of Glutamic Acid Using eq 2[a]

	GLU−	GLU0	GLU+	Cl^-	Na^+
P_{CPN} (Å/ps)	2.41×10^{-17}	1.26×10^{-4}	9.28×10^{-3}	1.28×10^{-14}	9.80×10^{-3}
P_{CPN} (cm/s)	2.41×10^{-13}	1.26	92.8	1.28×10^{-10}	98.0

[a]For comparison, those of Cl^- and Na^+ are also presented.

molecule between the two compartments divided by a lipid membrane, J can become meaningful when the concentration gradient between the two regions is significant despite a small value of P_m. Thus, the currents of GLU− observed in the experiments could be due to the concentration difference between the cis and trans sides of the bilayer used in the experiment. The second possibility is the substantial difference in the activity coefficient of NaGlu between the experiments and the MD simulations in this study. In the experiment the acquired activity coefficient for NaGlu was much lower than unity,[16] while the ion activity coefficient in the MD simulation is always assumed to be unity because there is only one GLU− in the simulation system. The third explanation could be an overestimation of the free energy barriers in the PMF calculations for GLU− due to artifacts in the MD simulation. This would be caused by the implementation of the finite size effect and periodicity and the use of a non-polarizable force field in the MD simulation method. A spurious destabilization of the ion in MD simulations has been discussed, and it was reported that corrections for the artifacts can reduce the energy barrier for ion transport through a gramicidin A channel by several kcal/mol.[57] The fourth possibility is that GLU− alters its protonation state from GLU− to GLU0 to avoid the high energy barriers and instability inside the CPNs because GLU0 has a much lower free energy barrier and is more stable than GLU− inside the channel, as shown in Figure 2. The transformation from GLU− to GLU0 can be made possible by GLU− taking a proton from neighboring water molecules before the anionic glutamic acid enters the CPN channel. The possibility of protonation state alteration in the glutamate transport through CPNs can be explored using ab initio MD (AIMD) or constant pH MD (pHMD) simulation methods.[58,59] Another possibility that the lipid head groups can interrupt the translocation of glutamic acid through the CPN and thus influence the PMF profiles for the acid was proposed during the review of this paper. To examine whether the lipid head groups are directly involved in the permeation of glutamic acid, we calculated the number density profiles of lipid head groups when a single GLU− is located near the channel entrance ($z = \pm 20$ Å). Figure S16 reveals that, due to the side chains of amino residues in the CPN, the lipid head groups maintain some distance away from the CPN and appear not to interrupt the translocation of GLU− as well as other forms of glutamic acid.

CONCLUDING REMARKS

In this study, we investigated the effect of the protonation state of glutamic acid on the energetics and dynamics of its transport through CPNs embedded in a DMPC bilayer using MD simulations combined with the ABF method. We evaluated permeability coefficients for the three protonation states of the acid on the basis of the solubility-diffusion model and clearly demonstrated that the translocation of glutamic acid strongly depends on its protonation state. Due to the cation-selective nature of the lumen of the CPNs, the cationic form of glutamic acid, GLU+, has the lowest free energy wells inside the CPN while the anionic form or glutamate, GLU−, exhibits the highest energy barriers. Small energy barriers and wells are observed inside the CPN for the zwitterionic form, GLU0. Although GLU− has repulsive interactions with CPNs, the anionic glutamic acid manages to develop favorable interactions with channel water through hydrogen bonding and rearrangement of water dipoles, which lowers energy barriers to some extent. The position-dependent diffusivity profiles reveal that, in contrast with the distinct diffusion coefficients in bulk water, all three protonation states of glutamic acid have similar frictional behavior inside the CPN due to confinement imposed by the lumen of CPNs. Unlike the considerable experimental values, the extremely small permeability coefficient for GLU− obtained from our MD simulations implies that a single glutamate is unlikely to pass through CPNs due to its huge energy barriers inside CPNs. There are several explanations for the discrepancy between our results and the experimental ones, including a large concentration gradient between the inside and outside of the vesicles and lipid bilayers in the experiments, the glutamate activity difference between the MD simulations and experiments, an overestimation of energy barriers due to artifacts in MD simulations, and/or an alteration in the protonation state from GLU− to GLU0 inside the channel to lower the barriers. We plan to examine the possibility of proton transfer from neighboring water molecules to glutamate permeating through CPNs by using advanced MD simulation methods such as AIMD or pHMD simulations. We believe that our present study will provide insight into the mechanism of action of glutamate translocation through glutamate transporters, as well as through CPNs. In addition, this work addresses the fundamental issue of whether the PMF provides a meaningful estimate of diffusion barriers in channels.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.3c02285.

Symmetrized PMF profiles with error bars for Cl^- and Na^+ across the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers; symmetrized total interaction energy (sum of electrostatic and van der Waals energy) profiles; symmetrized electrostatic interaction energy profiles with the system components for three protonation forms; symmetrized van der Waals interaction energy profiles with the system components for three protonation forms; symmetrized PMF decomposition profiles for three protonation forms; symmetrized total interaction energy (sum of electrostatic and van der Waals energy) profiles for Cl^- and Na^+; symmetrized electrostatic interaction energy profiles with the system components for Cl^- and Na^+; symmetrized van der Waals interaction energy profiles with the system components for Cl^- and Na^+; the number of hydrogen bonds between glutamic acid and water; the number of hydrogen bonds between glutamic acid and the CPN; RDF profiles for water hydrogen atoms around the nitrogen and oxygen atoms; RDF profiles for the amide hydrogen atoms of the CPN backbone around

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the nitrogen and oxygen atoms; position-dependent diffusion coefficient for three protonation forms of glutamic acids; position-dependent diffusion coefficient for Cl^- and Na^+; local resistivity, $R_{CPN}(z)$, profiles for Cl^- and Na^+ (PDF)

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Notes

The authors declare no competing financial interest.

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